

Markscheme

Mathematics: analysis and approaches
Practice question bank

137. (a)

The complex number z satisfies $|z - 4| = |z - 2i|$ and $\arg(z) = \frac{\pi}{4}$.

Find z in the form $a + bi$.

$|z - 4| = |z - 2i|$ is the perpendicular bisector of $(4, 0)$ and $(0, 2)$: $y = 2x - 3$ (M1) A1

$\arg(z) = \frac{\pi}{4}$ is the line $y = x$ (for $x > 0$) (M1)

solve simultaneously: $x = 2x - 3 \Rightarrow x = 3$ (M1)

$z = 3 + 3i$ A1

[6 marks]